PREFACE:

This project, "**Deep Fake Detection using Vision Transformers and Sequential Neural Networks on Diverse Datasets** " was undertaken as part of the academic requirements at the Symbiosis Institute of Geoinformatics. With advances in artificial intelligence and the technology of machine learning, deepfakes have risen as a challenge to the world of digital media, cybersecurity, and trust. Deepfakes—the use of sophisticated algorithms to create hyper-realistic but fabricated images, videos, or audio recordings—range in applications, from misinformation and identity theft to political manipulation and personal privacy invasions.

AIM :

This project was developed for deepfake detection that is strongly required in a methodical way. We worked with several datasets: the 140k Real and Fake Faces dataset and numerous other datasets with deepfake and authentic face images. This provided the training and testing of our detection models with a solid base and a diverse representative sample of deepfake and non-deepfake media.

The key models for detection are convolutional neural networks, thanks in part to the help of PyTorch libraries but also VIT Model. These provide effective performance in major domains, in particular in tasks related to image recognition, by spotting subtle inconsistencies and anomalies that distinguish real faces from synthesized ones. By exploiting the flexibility and dynamic computational capabilities of the Vision Transformer framework, the VIT model has greatly helped in devising experiments for complex and scalable deep learning.

COURSE OUTCOME :

Throughout this project, we have carried out meticulous data preprocessing towards enhancing model performance, hyperparameter fine-tuning to get the best accuracies, and model evaluation to ensure robust and reliable results. Our findings have shown great potential in high-accuracy deepfake detection, lending its significance to the fight against digital fraud and misinformation.